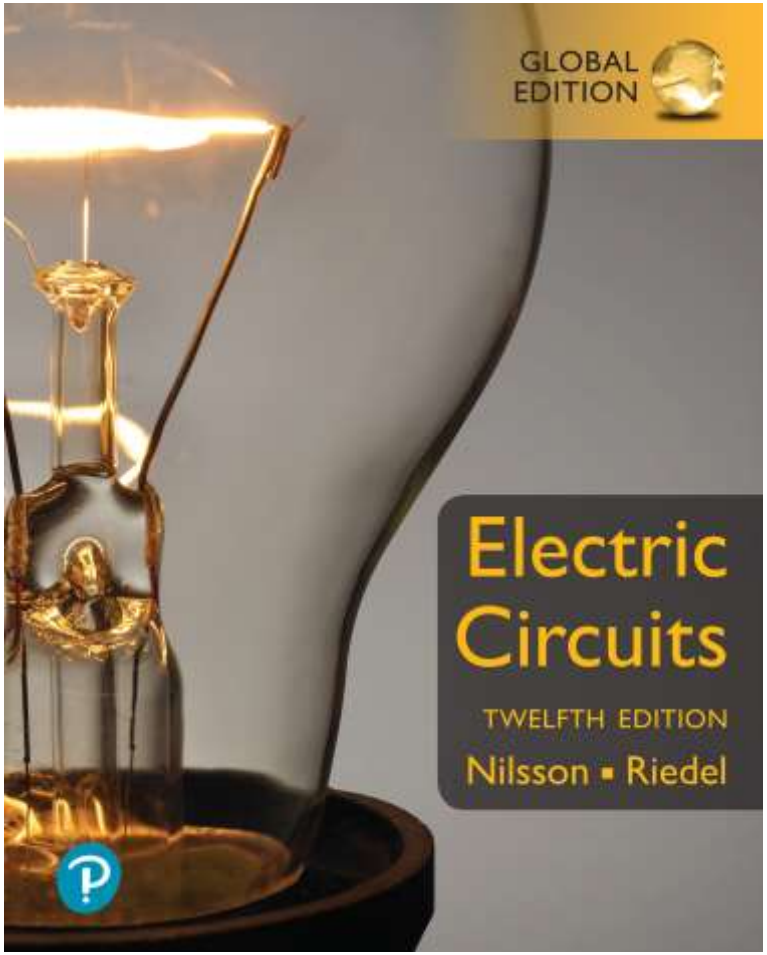
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 1

Techniques of Circuit Analysis

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Chapter 4 – techniques of circuit analysis (1 of 26)

Objectives:

- **4.1** Know how to use the **node voltage** method to solve a circuit
- **4.2** Know how to use the **mesh current method** to solve a circuit
- **4.3** Be able to choose the appropriate circuit analysis method to use for a particular circuit
- **4.4** Know how to use source transformation to simplify a circuit
- **4.5** Be able to calculate the Thevenin and Norton equivalents for a circuit
- **4.6** Understand and be able to use the condition for maximum power transfer to a load

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Chapter 4 – techniques of circuit analysis (2 of 26)

Terminology:

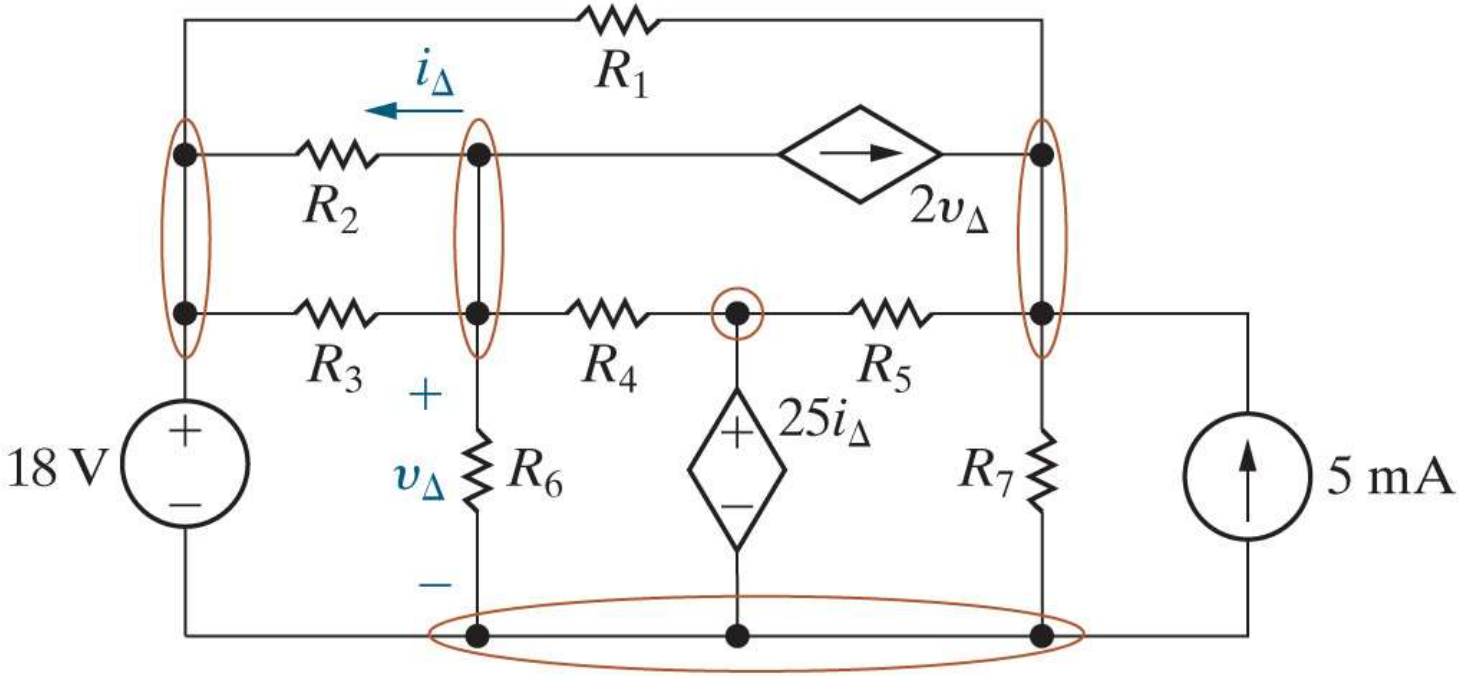
- Node – a point where two or more circuit elements are joined
- Essential node – a point where three or more circuit elements are joined
- Loop – a path whose ending node is the same as its starting node, and does not contain any intermediate nodes more than once
- Mesh – a loop that does not contain any other loops

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Chapter 4 – techniques of circuit analysis (3 of 26)

Identify all of the nodes in this circuit.

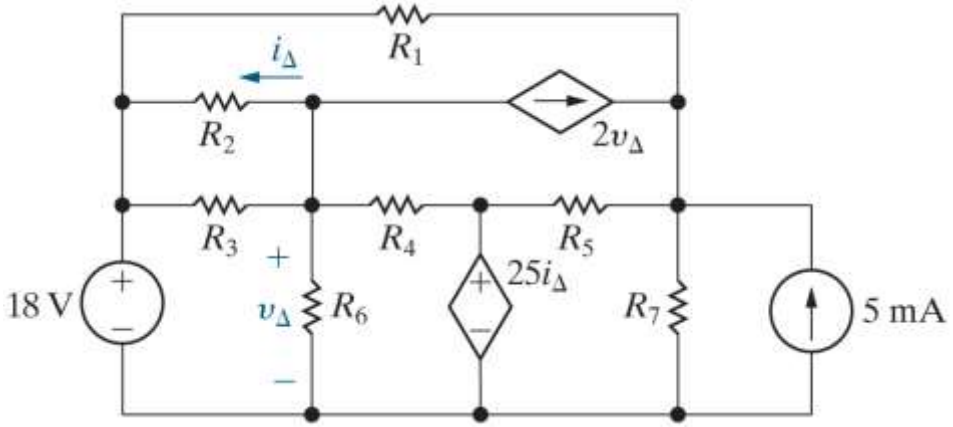


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Chapter 4 – techniques of circuit analysis (4 of 26)

How many essential nodes does this circuit have?



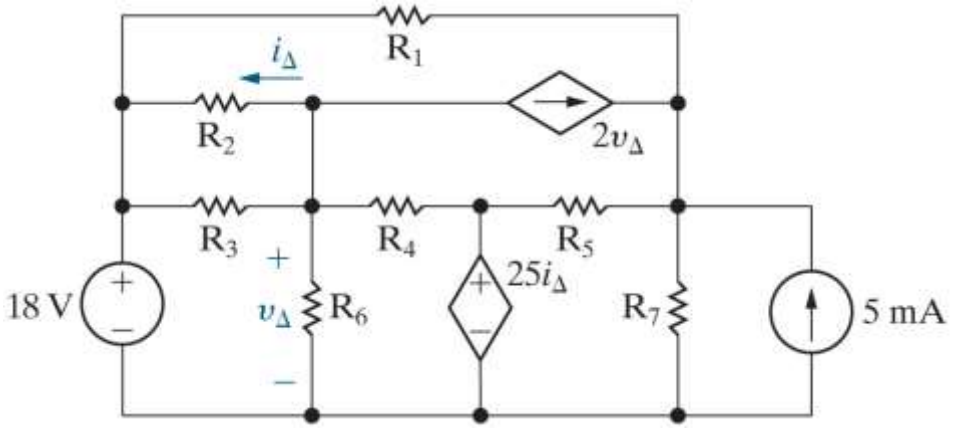
- ✗ A. 4
- ✓ B. 5
- ✗ C. 7
- ✗ D. 10

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Chapter 4 – techniques of circuit analysis (5 of 26)

Identify all of the loops in this circuit.



R_1, dep. source, R_2

R_1, R_5, R_4, R_2

Dep. Source, R_4, R_5

R_2, dep. source, R_5, R_4, R_3

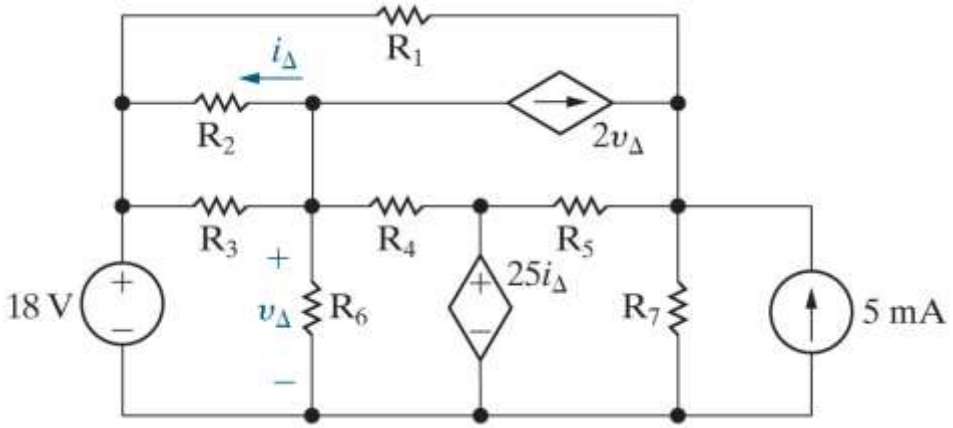
Etc . . .

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Chapter 4 – techniques of circuit analysis (6 of 26)

How many meshes does this circuit have?



- ✗ A. 4
- ✗ B. 5
- ✓ C. 7
- ✗ D. 10

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Chapter 4 – techniques of circuit analysis (7 of 26)

- Node voltage method:
  - Based on writing KCL equations at essential nodes
  - Solves for node voltages
  - Analysis Method 4.1

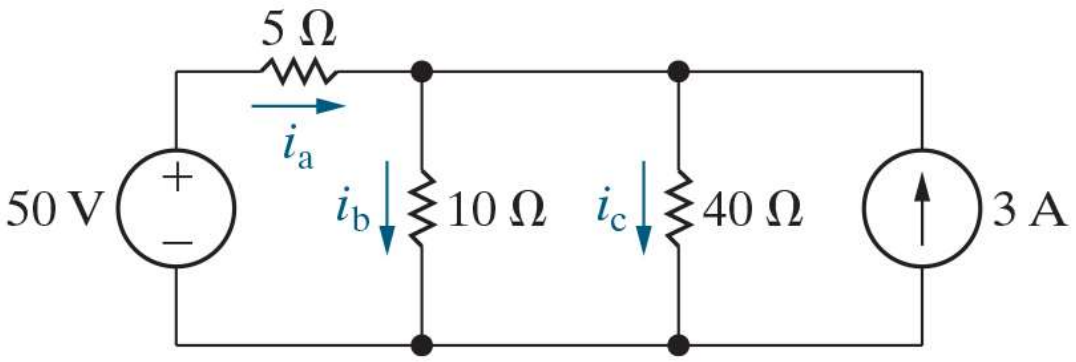
    1. Identify each essential node.
    2. Pick and label a reference node; then label the node voltages at the remaining essential nodes.
    3. Write a KCL equation for every nonreference essential node.
    4. Solve the equations to find the node-voltage values.
    5. Solve the circuit using node voltages from Step 4 to find component currents, voltages, and power values.

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Chapter 4 – techniques of circuit analysis (8 of 26)

Use the node voltage method to find the currents shown in this circuit.

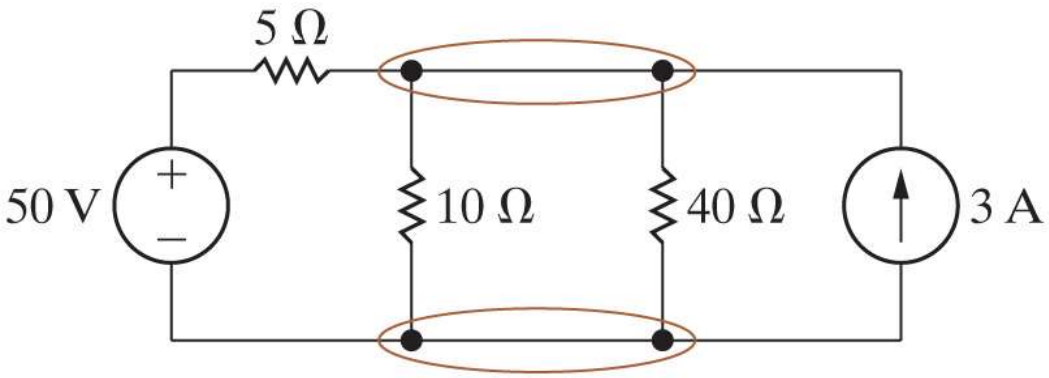


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Chapter 4 – techniques of circuit analysis (9 of 26)

- Node voltage method:
  - Step 1 – Identify each essential node.

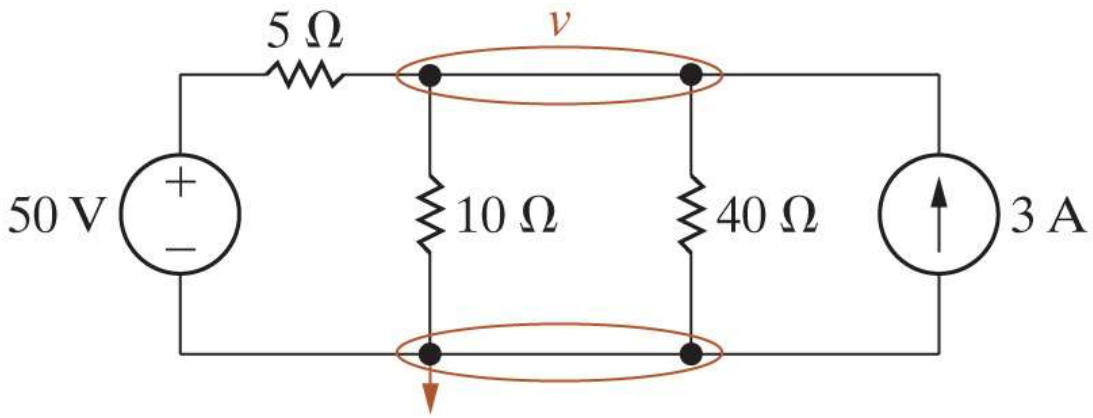


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Chapter 4 – techniques of circuit analysis (10 of 26)

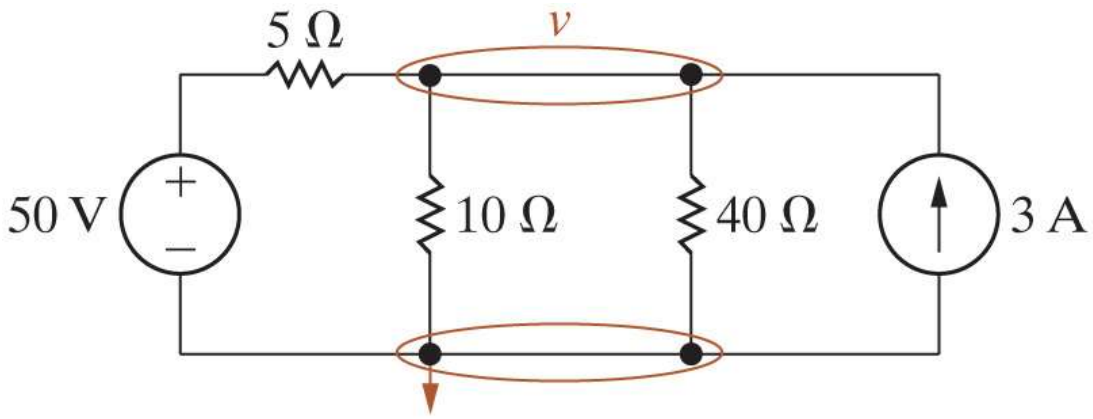
- Node voltage method:
  - Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.



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Chapter 4 – techniques of circuit analysis (11 of 26)

- Node voltage method:
  - Step 3 – Write a KCL equation for every nonreference essential node.



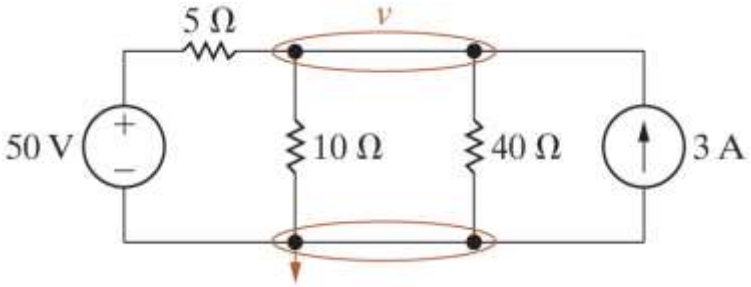
$$\text{At } v: \quad \frac{v - 50}{5} + \frac{v - 0}{10} + \frac{v - 0}{40} - 3 = 0$$

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Chapter 4 – techniques of circuit analysis (12 of 26)

- Node voltage method:
  - Step 4 – Solve the equations to find the node-voltage values.



$$\frac{v-50}{5}+\frac{v-0}{10}+\frac{v-0}{40}-3=0$$

$$\Rightarrow \quad v\left(\frac{1}{5}+\frac{1}{10}+\frac{1}{40}\right)=3+\frac{50}{5}$$

$$\Rightarrow \quad (40)\left[v\left(\frac{1}{5}+\frac{1}{10}+\frac{1}{40}\right)\right]=(40)\left[3+\frac{50}{5}\right]$$

$$\Rightarrow \quad v(8+4+1)=120+400$$

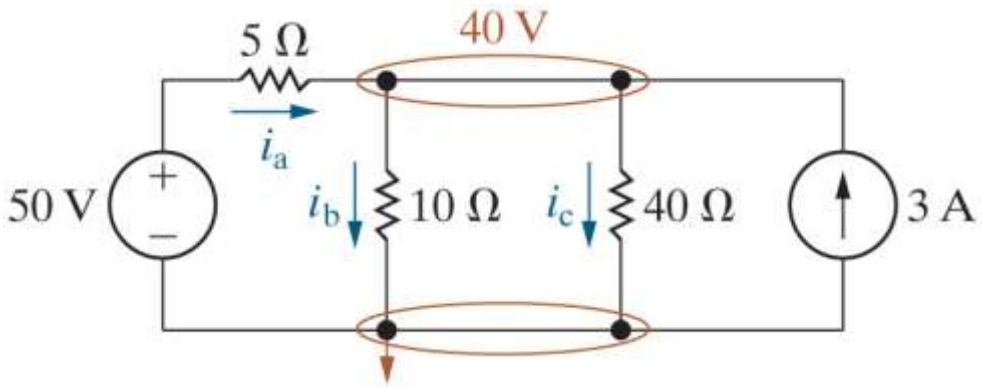
$$\Rightarrow \quad v=520/13=40\ \text{V}$$

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Chapter 4 – techniques of circuit analysis (13 of 26)

- Node voltage method:
  - Step 5 – Solve the circuit using node voltages from Step 4.



$$i_a = \frac{50 - 40}{5} = 2 \text{ A}$$

$$i_b = \frac{40 - 0}{10} = 4 \text{ A}$$

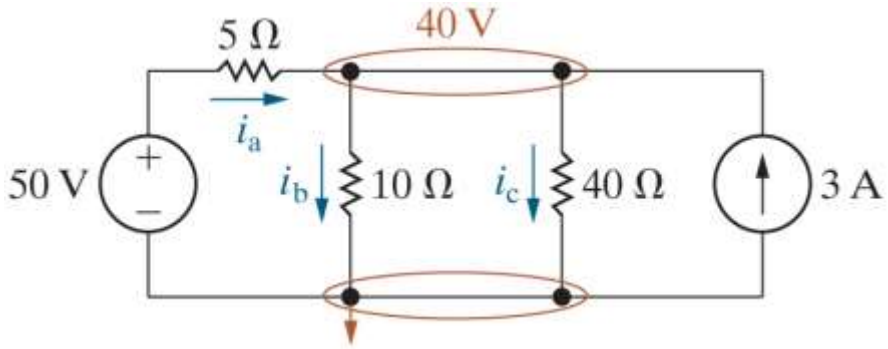
$$i_c = \frac{40 - 0}{40} = 1 \text{ A}$$

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Chapter 4 – techniques of circuit analysis (14 of 26)

We just showed that the voltage drop between the top node and the reference node in this circuit is 40 V. Find the power for the current source.



- ✗ A. 120 W
- ✗ B. −100 W
- ✓ C. −120 W
- ✗ D. 40 W

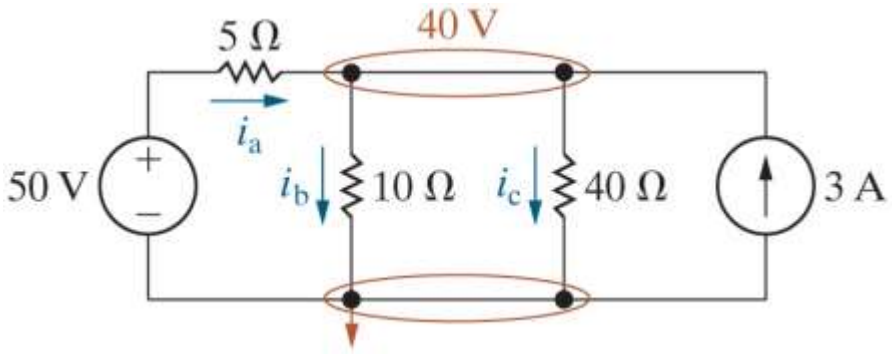
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Chapter 4 – techniques of circuit analysis (15 of 26)

Node voltage method:

- Check that the power balances:



$$i_a = \frac{50 - 40}{5} = 2 \text{ A}$$

$$i_b = \frac{40 - 0}{10} = 4 \text{ A}$$

$$i_c = \frac{40 - 0}{40} = 1 \text{ A}$$

$$p_{50V} = -(50)(2) = -100 \text{ W}$$

$$p_5 = 5(2)^2 = 20 \text{ W} \quad p_{10} = (40)^2/10 = 160 \text{ W}$$

$$p_{40} = (40)^2/40 = 40 \text{ W} \quad p_{3A} = -(40)(3) = -120 \text{ W}$$

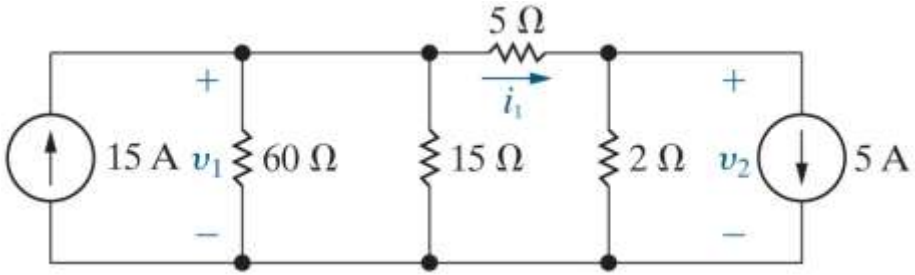
$$\sum p = -100 + 20 + 160 + 40 - 120 = 0$$

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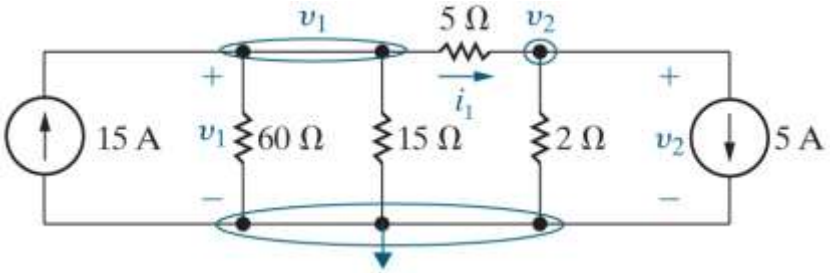
Chapter 4 – techniques of circuit analysis (16 of 26)

- Node voltage method:
  - Find the two voltages and the current indicated.



Step 1 – Identify each essential node.

Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.

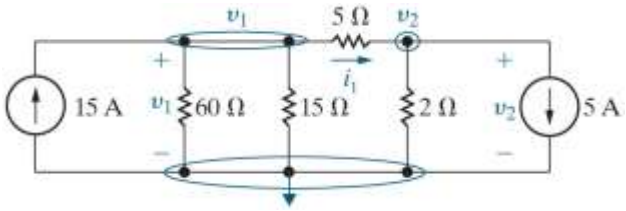


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Chapter 4 – techniques of circuit analysis (17 of 26)

- Node voltage method:
  - Step 3 – Write a KCL equation for every nonreference essential node.
  - Step 4 – Solve the equations to find the node-voltage values.



At v_1: $-15 + \frac{v_1}{60} + \frac{v_1}{15} + \frac{v_1 - v_2}{5} = 0$

At v_2: $\frac{v_2 - v_1}{5} + \frac{v_2}{2} + 5 = 0$

Standard form: $v_1\left(\frac{1}{60} + \frac{1}{15} + \frac{1}{5}\right) + v_2\left(-\frac{1}{5}\right) = 15$

$$v_1\left(-\frac{1}{5}\right) + v_2\left(\frac{1}{5} + \frac{1}{2}\right) = -5$$

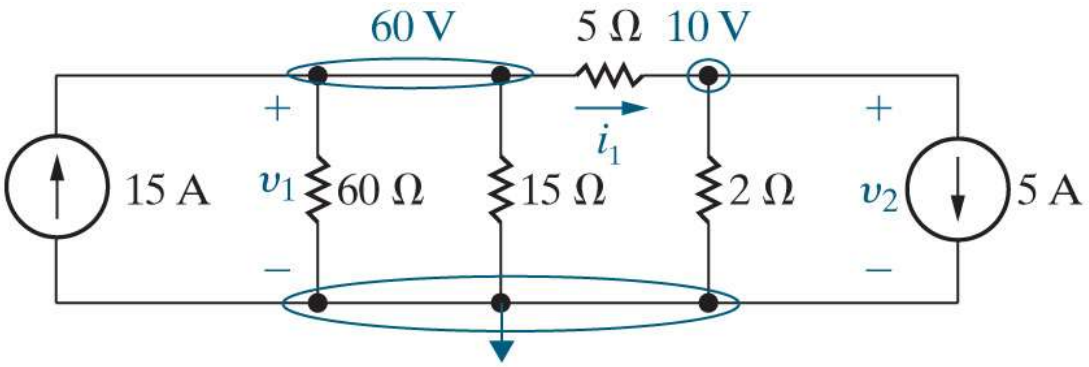
Calculator solution: $v_1 = 60$ V; $v_2 = 10$ V

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Chapter 4 – techniques of circuit analysis (18 of 26)

- Node voltage method (continued):
  - Step 5 – Solve the circuit using node voltages from Step 4.



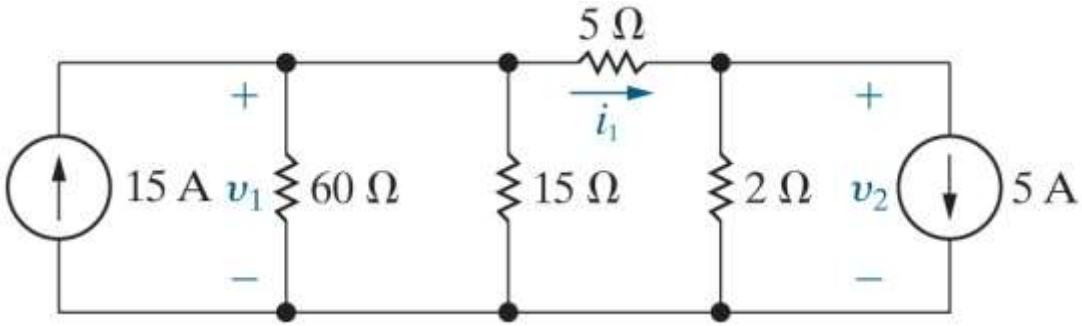
Calculate i_1 : $i_1 = (60 - 10)/5 = 10$ A

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Chapter 4 – techniques of circuit analysis (19 of 26)

If v_1 is 60 V and v_2 is 10 V, what is the power associated with the 60 Ω resistor?



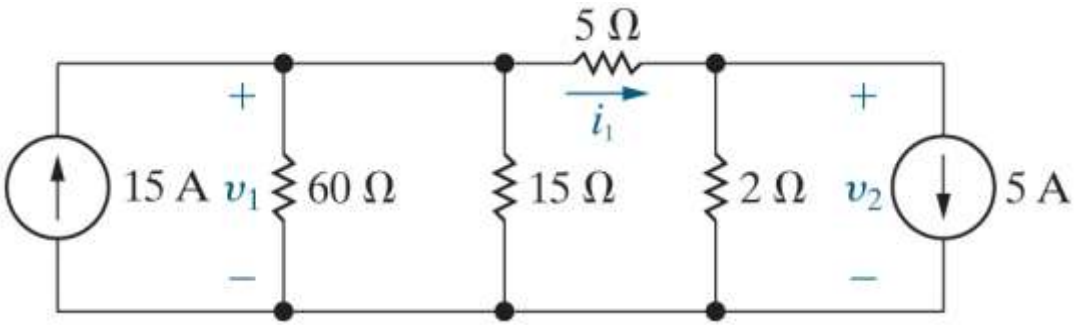
- ✔ A. 60 W
- ✘ B. 240 W
- ✘ C. 3600 W
- ✘ D. None of the above

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Chapter 4 – techniques of circuit analysis (20 of 26)

If v_1 is 60 V and v_2 is 10 V, what is the power associated with the 5 A source?



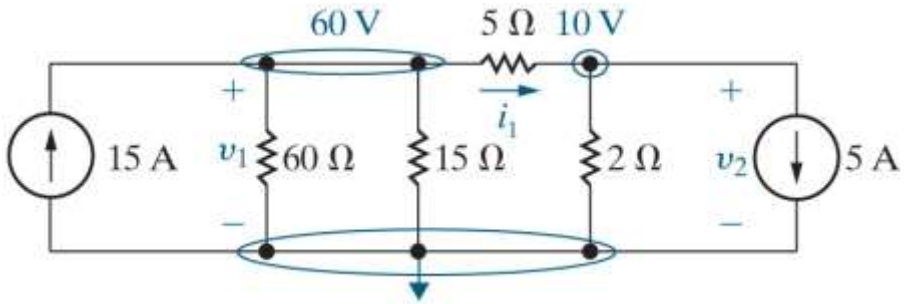
- ✗ A. 900 W
- ✗ B. −900 W
- ✗ C. −50 W
- ✓ D. 50 W

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Chapter 4 – techniques of circuit analysis (21 of 26)

- Node voltage method (continued):
  - Check that the power balances



$$p_{15A} = -(15)(60) = -900 \text{ W}; \quad p_{60} = 60^2/60 = 60 \text{ W};$$

$$p_{15} = 60^2/15 = 240 \text{ W}; \quad p_5 = (60-10)^2/5 = 500 \text{ W};$$

$$p_2 = 10^2/2 = 50 \text{ W}; \quad p_{5A} = (5)(10) = 50 \text{ W}$$

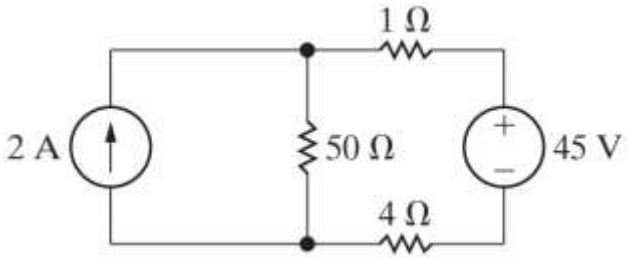
$$\sum p = -900 + 60 + 240 + 500 + 50 + 50 = 0$$

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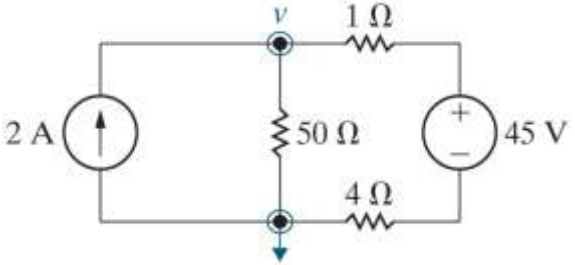
Chapter 4 – techniques of circuit analysis (22 of 26)

- Node voltage method:
  - Find the power associated with the current source.



Step 1 – Identify each essential node.

Step 2 – Pick and label a reference node; then label the node voltages at the remaining essential nodes.

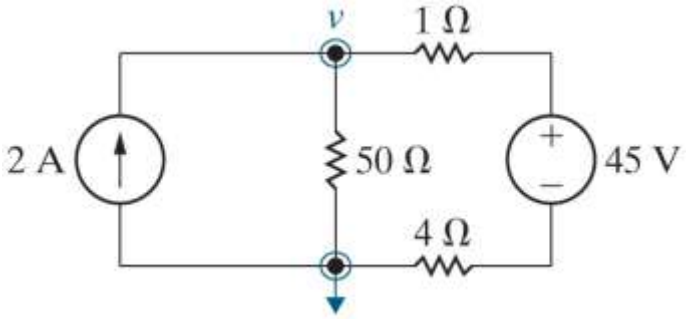


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Chapter 4 – techniques of circuit analysis (23 of 26)

- Node voltage method:
  - Step 3 – Write a KCL equation for every nonreference essential node.
  - Step 4 – Solve the equations to find the node-voltage values.



KCL at v: $-2 + \frac{v}{50} + \frac{v - 45}{1 + 4} = 0$

Standard form: $v\left(\frac{1}{50} + \frac{1}{5}\right) = 2 + \frac{45}{5}$

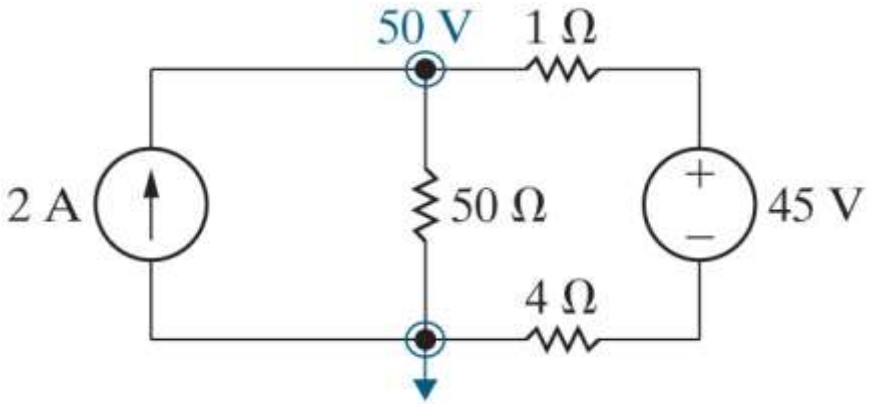
Solving: (50) $\left[v\left(\frac{1}{50} + \frac{1}{5}\right) = 2 + \frac{45}{5}\right] \Rightarrow v(1 + 10) = 100 + 450 \therefore v = \frac{550}{11} = 50 \text{ V}$

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Chapter 4 – techniques of circuit analysis (24 of 26)

- Node voltage method:
  - Step 5 – Solve the circuit using node voltages from Step 4.



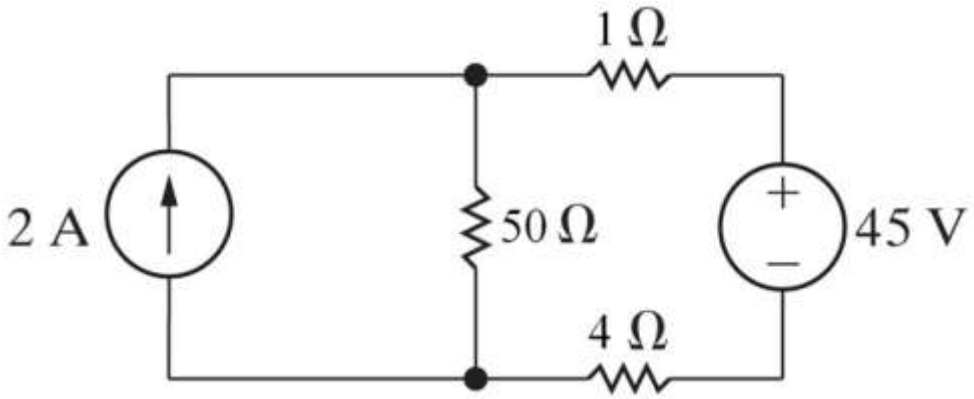
Power for the current source: $p_{2A} = -(50)(2) = -100$ W

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Chapter 4 – techniques of circuit analysis (25 of 26)

The current source is absorbing power.



✗ A. True

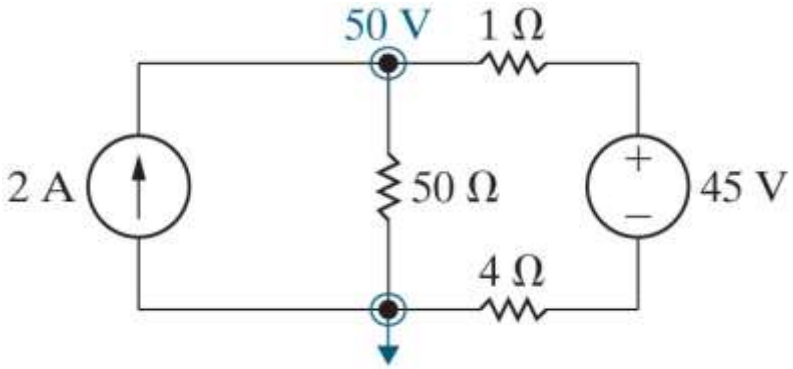
✓ B. False

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Chapter 4 – techniques of circuit analysis (26 of 26)

Power balance:



Component	Equation	p [W]
2 A	$-(2)(50)$	−100
45 V	$(45)[(50-45)/5]$	45
1 Ω	$(1)^2(1)$	1
4 Ω	$(1)^2(4)$	4
50 Ω	$(50)^2/50$	50

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